



Pixel 2 XL taken apart

iFixit has taken apart Google's latest Pixel 2 XL to reveal the Snapdragon 835 with Google's custom designed SoC. <http://dgit.in/Pixel2TDwn>



Apple sued for new emojis

Emonster, a Tokyo-based company that owns the trademark for Animojis (Animal Emojis) is suing Apple. <http://dgit.in/appleanimoji>

POWER AND TEMPERATURE

RADEON RX VEGA 64

When idle, the Radeon RX Vega 64 consumes approximately 85 Watts, lower than its predecessor, the R9 Fury X which used to consume approximately 96 Watts under load. Under load using FurMark, we see the RX Vega 64 eat up around 405 Watts which is quite a lot, yet lower than the R9 Fury X. Our FurMark settings are a bit conservative, so we switched to video games for a power test. It was surprising to see the RX Vega 64 consume even more power, going up to 460 Watts at peak load. We also looked at the temperatures while running these tests. Under idle conditions, it would be around 33-35 degrees Celsius while under load it would go to 83-85 degrees Celsius.

NVIDIA GTX 1080

The GTX 1080 FE has a lower power draw under all conditions. The 1080 Ti, on the other hand, is quite similar to the Vega 64 with regards to draw. Coming back to the GTX 1080, under idle conditions the GPU would be around 31-33 degrees Celsius while under Furmark, it would rise to 82-84 degrees Celsius. When checking power draw, the 1080 Founders Edition would consume approximately 82 Watts at idle and in Furmark, that would go all the way up to about 300 watts. In our gaming benchmarks, the power draw did increase a little bit to about 315 Watts. Compared to its predecessor, i.e. the GTX 980, the GTX 1080 consumes more power under load and idle conditions.

GAMING PERFORMANCE

RADEON RX VEGA 64

Here's the FPS table for some of the games that we benchmarked on the RX Vega 64:

DOOM (4K) - 69 FPS,
DOOM (FHD) - 184 FPS,
Ashes of the Singularity (4K) - 59 FPS,
Ashes of the Singularity (FHD) - 81 FPS,
Total War: Warhammer (4K) - 49 FPS,
Total War: Warhammer (FHD) - 94 FPS,

GTA V (4K) - 29 FPS,
GTA V (FHD) - 80 FPS,
Dawn of War III (4K) - 48 FPS,
Dawn of War III (FHD) - 83 FPS,
Deus Ex: MD (4K) - 29 FPS,
Deus Ex: MD (FHD) - 77 FPS.

NVIDIA GTX 1080

Subsequently, the same on the NVIDIA GTX 1080 Founders Edition:

DOOM (4K) - 62 FPS,
DOOM (FHD) - 170 FPS,
Ashes of the Singularity (4K) - 57 FPS,
Ashes of the Singularity (FHD) - 85 FPS,
Total War: Warhammer (4K) - 50 FPS,
Total War: Warhammer (FHD) - 121 FPS,

GTA V (4K) - 38 FPS,
GTA V (FHD) - 110 FPS,
Dawn of War III (4K) - 44 FPS,
Dawn of War III (FHD) - 98 FPS,
Deus Ex: MD (4K) - 28 FPS,
Deus Ex: MD (FHD) - 75 FPS.

FRAME-TIME ANALYSIS

RADEON RX VEGA 64

When we look at frame times of Rise of the Tomb Raider using FRAPs, we see that they are more or less equally matched. We then switched to The Witcher 3 and scaled the resolution to 4K. We did notice a couple of spikes in the frame time charts indicating anomalous frame rates. We then used HITMAN to check for spikes and noticed next to none except a few towards the end of our benchmark runs. The last game we ran was GTA V which had quite a few closely packed spikes towards the latter part of the built-in benchmark run, however, visually, we couldn't see that big a difference.

NVIDIA GTX 1080

We noticed equally anomalous spikes on the GTX 1080 Founders Edition as well, however, they were synchronous with scene changes so these are normal. However, we did notice fewer spikes in the frame times with the GTX 1080. The GTX 1080 Ti worked smoothly with HITMAN as well, at both, FHD and 2K resolutions. In GTA V, the last benchmark that we ran, we noticed a similar set of spikes as we did with the RX Vega 64, however, the amplitude of the spikes were lower than the RX Vega 64. Overall, the two cards were fairly consistent with few spikes to worry about.

CONCLUSION

AMD has produced quite a great product with the RX Vega 64 for enthusiasts. It does perform better in a few games such as DOOM and Ashes of the Singularity, while the NVIDIA GTX 1080 Founders Edition does better in most other games. Especially, when you scale down on the resolution and the 1080 FE gives comparatively greater FPS numbers. Overall, we're looking at close gaming performance with the two graphics cards. In power consumption, AMD has done a great job at improving the architecture from its

predecessor, however, compared to the competition, the Vega 64 is a power hog. Its temperatures aren't that better off either. At the end, we're looking at two similarly-priced graphics cards with very similar performance but vastly different power draws. Extrapolating electricity costs over a longer time, the RX Vega 64 can be more expensive to run which works in favour of the NVIDIA GTX 1080. However, this is the first generation of the Vega architecture and we hope to see better with the second generation. Till then, it'll have to be the NVIDIA GTX 1080 Founders Edition. **[4]**